Athernex

A TRI COLLEGE TECH-FEST
BY DSCE, BMSCE & RVU

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Team Check-in

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Problem Statement

The freelance economy is massive ($1.5T+, 76M+ workers), yet trust remains a core bottleneck, slowing down transactions and long-term collaboration between clients and freelancers

Freelancers frequently face payment insecurity, including delayed payments and non-payment, along with high escrow fees (5–10%) and long settlement delays (3–5 days) that impact cash flow

Freelancers are heavily affected by platform lock-in, where reputation and ratings built on one platform (e.g., Upwork, Fiverr) cannot be transferred elsewhere, limiting growth and independence

Clients struggle with quality verification, as freelancer claims are difficult to validate without platform-controlled reviews, leading to uncertainty and hiring risks

Both clients and freelancers face inefficient dispute resolution systems, where processes are slow, biased, and often place the burden of proof unfairly on freelancers

Current systems suffer from siloed reputation, binary trust models, and privacy trade-offs, where users must expose sensitive work or transaction history but still receive only limited, non-nuanced trust signals

Solution

Introduces a reputation-weighted escrow system, where trust directly impacts transaction terms and reduces inefficiencies

Reputation-Backed Staking: Freelancers with proven track records can lock less collateral or pay lower escrow fees, reducing financial friction

Verifiable Completion Credentials: Every successful project generates cryptographically signed credentials, building a portable and tamper-proof reputation

Transparent Dispute Flow: Implements rule-based, time-bound dispute resolution, ensuring fairness and reducing ambiguity for both parties

Cross-Platform Portability: Uses W3C Verifiable Credential standards, enabling seamless verification across platforms without vendor lock-in

Solution

Traditional platforms have closed reputation systems, while SkillBond enables portable, cross-platform reputation

Fiverr/Upwork rely on centralized reviews, whereas SkillBond uses cryptographically verifiable credentials (trustless verification)

Existing platforms charge high escrow fees and impose delays, while SkillBond reduces cost via reputation-weighted incentives

Dispute resolution today is platform-controlled and often opaque; SkillBond introduces transparent, rule-based mechanisms

Current systems offer binary trust (ratings), while SkillBond provides multi-dimensional, verifiable reputation signals

Platforms act as intermediaries controlling trust, whereas SkillBond acts as a protocol layer enabling trust across ecosystems

Features

Reputation-Weighted Escrow
Escrow terms dynamically adjust based on freelancer reputation, enabling lower fees and reduced collateral for proven performers

Verifiable Completion Credentials
Each completed project generates a cryptographically signed credential, creating a tamper-proof and portable reputation trail

Cross-Platform Reputation Portability
Built on W3C Verifiable Credential standards, allowing freelancers to carry trust across platforms without being locked in

Transparent & Rule-Based Dispute System
Time-bound, clearly defined dispute resolution reduces ambiguity and ensures fairness for both clients and freelancers

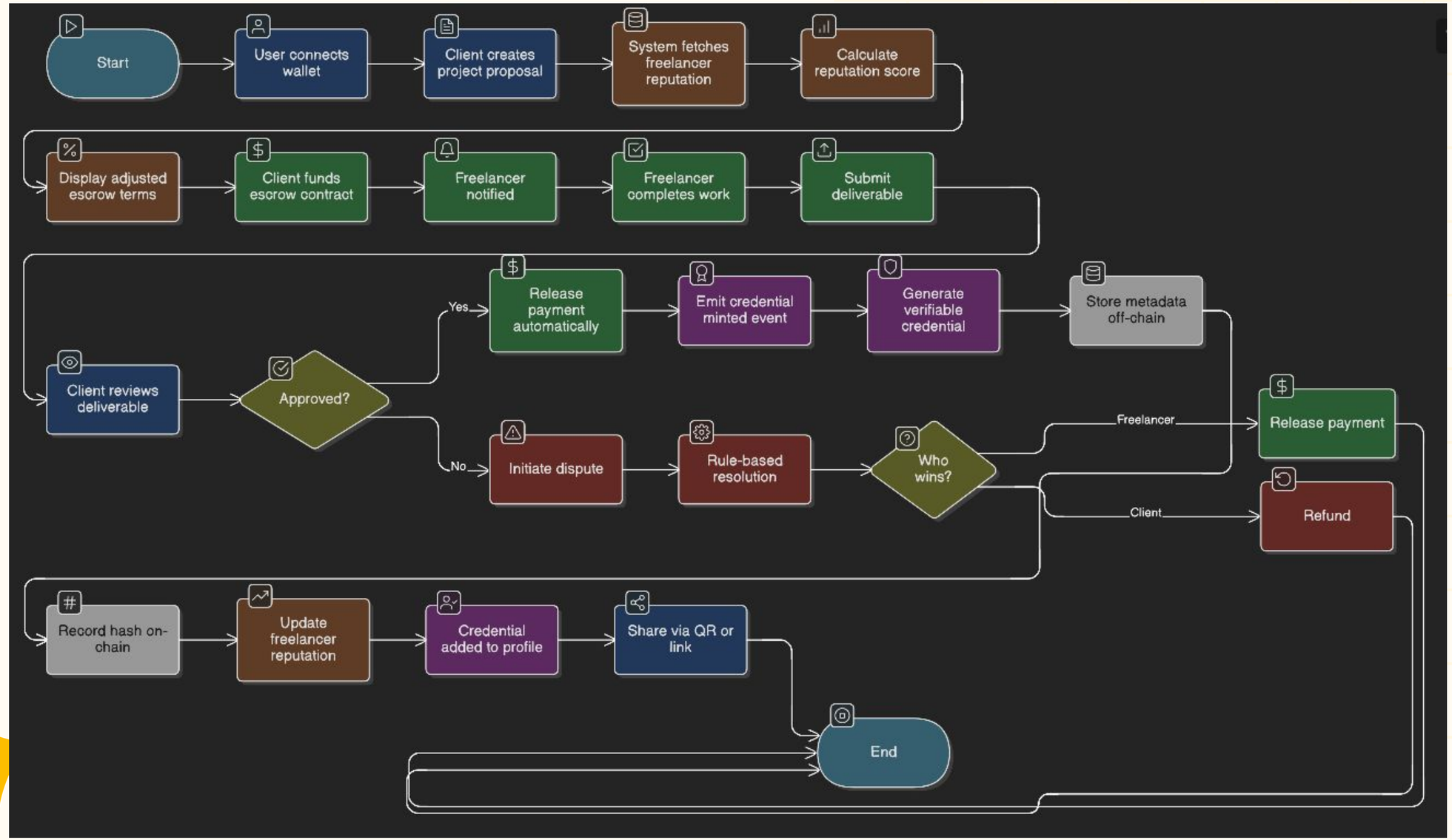
Decentralized Reputation Scoring Engine
Reputation is computed using a weighted model (completion rate, recency, issuer credibility) instead of simple ratings

On-Chain + Off-Chain Hybrid Architecture
Smart contracts handle escrow and credential hashes, while metadata is stored off-chain for scalability and efficiency

Privacy-Preserving Credential Sharing
Users can share proof of work via QR codes or links without exposing full transaction history

3-4 Main Features

Flow Chart


Start
User connects wallet
Client creates project proposal
System fetches freelancer reputation
Calculate reputation score
Display adjusted escrow terms
Client funds escrow contract
Freelancer notified
Freelancer completes work
Submit deliverable
Client reviews deliverable
Approved?
Yes
No
Release payment automatically
Emit credential minted event
Generate verifiable credential
Store metadata off-chain
Initiate dispute
Rule-based resolution
Who wins?
Freelancer
Client
Release payment
Refund
Record hash on-chain
Update freelancer reputation
Credential added to profile
Share via QR or link
End

Tech Stack

Frontend
- React.js / Next.js
- Tailwind CSS
- Wallet Integration (MetaMask / WalletConnect)

Blockchain Layer
- Solidity Smart Contracts
- Polygon / Base Testnet
- ethers.js / Wagmi

Backend & APIs
- Node.js + Express
- REST API / GraphQL
- Event Listener (Contract Events)

Storage Layer
- IPFS (via Pinata)
- Off-chain JSON Storage
- On-chain Hash Storage

Identity & Credentials
- W3C Verifiable Credentials (VC)
- Decentralized Identifiers (DIDs)
- Cryptographic Signatures (ECDSA)

Core Logic
- Reputation Scoring Engine
- Weighted Algorithm (Completion + Recency + Credibility)
- Dispute Resolution Logic

Deployment & Tools
- Vercel (Frontend)
- Render / Railway (Backend)
- Hardhat / Foundry
- GitHub

Impact

HOW DOES YOUR IDEA CREATE VALUE?

Freelancers gain greater financial security and predictability, as payments are protected through automated escrow systems, reducing dependence on platform policies and minimizing risks of non-payment

Clients benefit from higher trust and transparency in hiring, with access to verifiable proof of a freelancer's past work and performance rather than relying on subjective reviews

Both parties experience lower costs and smoother transactions, as reputation-based incentives reduce unnecessary fees and streamline the overall engagement process

A more balanced and fair dispute environment emerges, where decisions are guided by predefined rules and verifiable data instead of opaque, platform-controlled judgments

From a broader ecosystem perspective, talent mobility improves significantly, allowing skilled freelancers to move freely across platforms without losing their reputation or starting from scratch

For regulators and governments, the system introduces greater accountability and auditability, as transactions and credentials are verifiable, traceable, and less prone to manipulation while still preserving user privacy

Atherne
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Thank You So Much

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